

MATH I – Week 1

Lecture presentation – PART B

on Number Systems

Numeral (numeration) systema, number sets, number systems

I. Distinguish between numeral systems, number sets and number systems

The terms **numeral systems**, **number sets**, and **number systems** refer to related but different ideas.

A **numeral system** is a method for writing or representing numbers using particular symbols and rules. Examples include the decimal system base 10, binary system base 2, hexadecimal system base 16, and Roman numerals. Thus, the same number can have different representations:

$$13_{10} = 1101_2 = D_{16} = \text{XIII}.$$

A **number set** is a collection of numbers having specified properties. The standard number sets include the natural numbers $\mathbb{N}$, integers $\mathbb{Z}$, rational numbers $\mathbb{Q}$, real numbers $\mathbb{R}$, and complex numbers $\mathbb{C}$, with

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}.$$

The term **number system** is broader. It can refer to a number set together with the operations and rules defined on it—for example, the real number system includes $\mathbb{R}$, arithmetic operations, ordering, and algebraic properties.

However, “number system” is sometimes also used informally to mean a numeral system. To avoid ambiguity, it is best to use **numeral system** for binary, decimal, and hexadecimal representations, and **number sets** or **number domains** for $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$

Why do we need numbers?

1. Why do we need numbers?

Suppose an engineer asks

- How long is this bridge?
- What is the voltage?
- How much energy is stored?
- How fast is the aircraft?

All these questions require numbers.

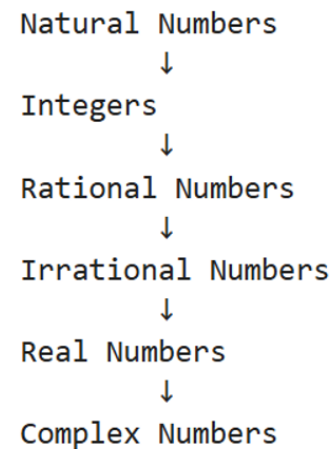
Numbers allow us to

- measure,
- compare,
- calculate,
- predict,
- optimize.

Engineering is impossible without numbers!

2. The evolution of numbers

Human civilization gradually invented new kinds of numbers whenever old ones became insufficient.



Every new system solved an important problem!

What kind of numbers do number systems represent?

Set Name	Set Symbol	Set Description
Natural Numbers	$\mathbb{N}$	The counting numbers
Whole Numbers		The counting numbers and 0
Integers	$\mathbb{Z}$	The natural numbers, their negatives, and 0
Rational Numbers	$\mathbb{Q}$	Fractions of integers
Irrational Numbers	$\mathbb{P}$	Numbers that cannot be written as a fraction of integers
Real Numbers	$\mathbb{R}$	The union of the rational and irrational numbers, all possible physical lengths, and their negatives

$\mathbb{C}$ — Complex numbers

$3 + 2i, -4i$

$\mathbb{R}$ — Real numbers

$\pi, \sqrt{2}$

$\mathbb{Q}$ — Rational numbers

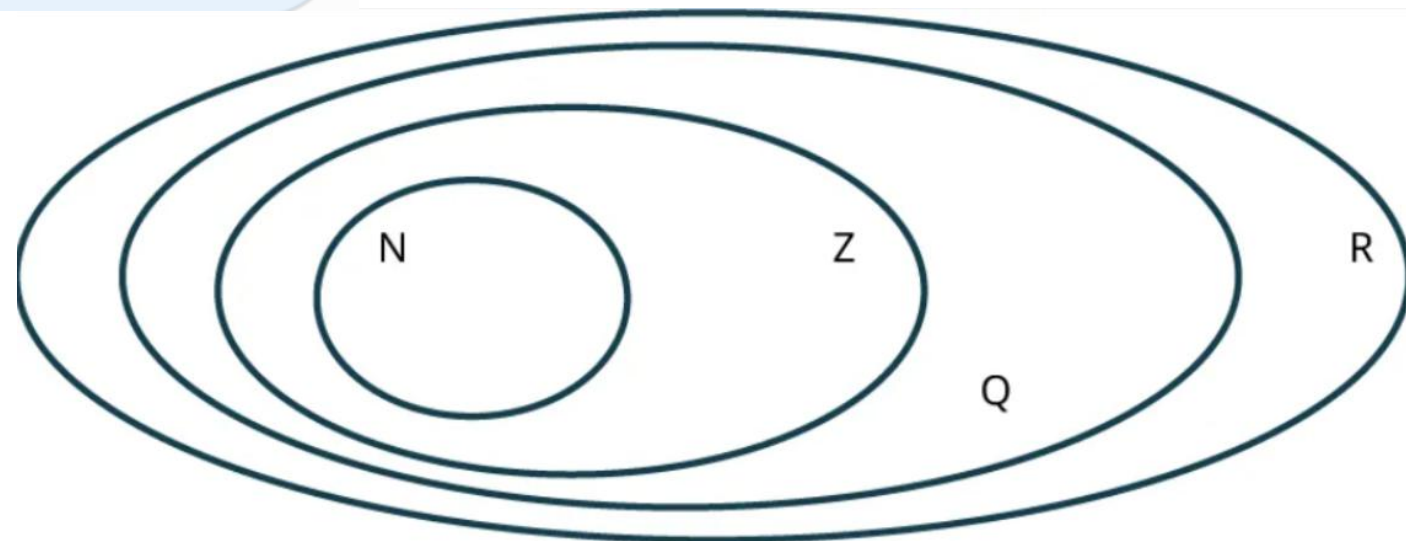
$2/3$

$\mathbb{Z}$ — Integers

-2

$\mathbb{N}$ — Natural numbers

Nested (Venn type)
diagrams



$\mathbb{R} \setminus \mathbb{Q}$ or $\mathbb{R} - \mathbb{Q}$

(for the irrational numbers)

A note about notation in set difference:

For two sets A and B ,

$$A \setminus B$$

means “the elements belonging to A but not to B ”:

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}.$$

As another example,

$$\mathbb{R} \setminus \mathbb{Q},$$

is the set of irrational numbers.

$$\mathbb{R} \setminus \mathbb{Q} = \{x \in \mathbb{R} : x \notin \mathbb{Q}\}.$$

Natural numbers and Integers

3. Natural Numbers

The first numbers humans used were

$1, 2, 3, 4, 5, \dots$

They answer

“How many?”

Examples

- 5 sensors
- 20 bolts
- 8 robots

Natural numbers are used for counting.

Notation: $\mathbb{N}$

4. Integers

Natural numbers cannot answer

What is $5 - 8$?

The answer is -3 .

Negative numbers were invented.

Integers are

$\dots, -3, -2, -1, 0, 1, 2, 3, \dots$

Notation: $\mathbb{Z}$

Examples

- temperature: -15°C
- depth: -50m

Rational numbers (fractions)

5. Rational Numbers

Integers cannot solve

$$1 \div 2 (= \frac{1}{2})$$

We need fractions.

Examples

$$\frac{1}{2}, \frac{3}{5}, -\frac{7}{4}$$

Notation: $\mathbb{Q}$

Every rational number can be written as

$$\frac{a}{b}$$

where

- a is an integer
- $b \neq 0$

Examples

$$0.25 = \frac{1}{4}$$

$$0.333333 \dots = \frac{1}{3}$$

Rational numbers have

- terminating decimals

or

- repeating decimals.

Irrational numbers

6. A surprising discovery

The ancient Greeks believed everything could be expressed as a fraction.

Then they studied a square. Suppose the side length equals 1.

Using Pythagoras, the diagonal equals $\sqrt{2}$.

Can $\sqrt{2}$ be written as $\frac{a}{b}$?

The answer is No. This was shocking.

7. Irrational Numbers

Irrational numbers cannot be written as fractions.

Examples: $\sqrt{2}$, $\sqrt{3}$, π , e

Their decimal expansions never end and never repeat.

Example: $\pi = 3.1415926535\dots$

Real numbers

8. Real Numbers

Real numbers combine

- rational numbers

and

- irrational numbers.

Notation: $\mathbb{R}$

Everything on the continuous number line is a real number.



-3 -2 -1 0 1 2 3

Every point corresponds to one real number.

Real numbers represent

- length
- time
- mass
- voltage
- pressure
- temperature
- velocity

Almost every engineering measurement is a real number.

Why real numbers are important?

9. Why are real numbers important?

Calculus studies continuous change. Suppose

- temperature changes continuously.
- Position changes continuously.
- Voltage changes continuously.
- Current changes continuously.

These quantities require "continuous" numbers.

The real numbers make continuity possible!

10. Density of the real numbers

Between any two real numbers there are infinitely many others.

Example between 1 and 2 we find 1.5,

between 1 and 1.5 we find 1.25,

between 1 and 1.25 we find 1.125, and so on.

This process never ends. ***Real numbers are infinitely dense.***

Properties of real numbers

12. The properties of real numbers

Real numbers satisfy important rules.

Commutative Law

Addition

$$a + b = b + a$$

Multiplication

$$ab = ba$$

Associative Law

$$(a + b) + c = a + (b + c)$$

$$(ab)c = a(bc)$$

Distributive Law

$$a(b + c) = ab + ac$$

Identity Elements

For addition: $a + 0 = a$

For multiplication: $a \cdot 1 = a$

Inverse Elements

For addition: $a + (-a) = 0$

For multiplication:

$$a \cdot \frac{1}{a} = 1$$

for

$$a \neq 0.$$

13. Ordering real numbers

Real numbers can be compared.

Examples

$$2 < 5$$

$$-3 < 1$$

Ordering allows us to define

- maximum
- minimum
- intervals

which become essential in calculus.

14. Intervals

Examples

Closed interval

$$[a, b]$$

contains both endpoints.

Open interval

$$(a, b)$$

does not contain the endpoints.

Half-open intervals

$$[a, b)$$

$$(a, b]$$

These appear throughout calculus.

Computer representation of numbers

(a very first discussion to highlight the issue)

15. An important point: Real numbers and computers

Here comes the surprise. Computers do **not** store real numbers. Instead, they store approximations.

Example (in Python)

```
0.1
```

is actually stored approximately.

Therefore

```
0.1 + 0.2
```

returns

```
0.30000000000000004
```

This is not a bug. It is a consequence of binary floating-point arithmetic. But this is something to discuss later in your studies.

In-Class activity

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Try to solve the following problems. Keep your solutions and check them with the answers.

The in-class activity is to facilitate your learning; it will not be graded.

Exercise 1: Set diagram

Draw a nested-set diagram representing

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}.$$

Place each of the following numbers in the correct region:

$$2026, \quad -9, \quad \frac{2}{3}, \quad \sqrt{10}, \quad \pi, \quad -5i, \quad -2 + 2i.$$

In-Class activity (cont'd)

Exercise 2: Rational or irrational?

Determine whether each number is rational or irrational. Justify your answer.

a. $\sqrt{49}$,

b. $\sqrt{18}$,

c. $3 + \sqrt{2}$,

d. $2\sqrt{5}$,

e. $\sqrt{2} + \sqrt{8}$,

f. $(\sqrt{5})^2$